RP2350B

Raspberry Pi
RP2350B QFN-80

U1
RP2350_80QFN

IO_D1..IO_D8 must be on consecutive pins to simplify DMA sampling

place R10 and R11 close to RP2350

Analog Inputs

J7
Conn_02x04_Counter_Clockwise

U2A U2B U2C U2D
TLV9004IPWR

D3 D4 D5 D6
BAS70-04

U2E
TLV9004IPWR

fg=159kHz

USB Input

J1
USB_C_Receptacle_USB2.0

F1
500mA

U5
USBLC6-2P6

U6
AP2112K-3.3

Variable Voltage Regulator

U10
MP28167-A

R_fb_top / R_fb_bot = V_max / V_ref - 1
= 5.16V / 1.637V - 1 = 2.15

Output Rails

U16 U17 U18
LM66100DCK

U4 U8 U11
INA180A2

fg=159Hz

R_shunt = V_adc / gain / I_max =
3V / 50 / 2A = 30mOhm

J4
Conn_02x04_Counter_Clockwise

Flash

U3
IS25LP064D-JKLA3-TR

Alternative: W25Q64JVZPIQ

LED

D1
1N4148WS

D2
SK6812MINI

Digital GPIO

U13 U14 U19 U20 U21 U22 U23 U24
SN74LVC1T45DBV

Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12 Q13 Q14 Q15 Q16 Q17 Q18
DMG2301L
BSS138

U15 U25
SRV05-4

J6
Conn_02x04_Counter_Clockwise

Buses

U26 U28 U30 U31 U32 U33
SN74LVC1T45DBV

U27 U29
SRV05-4

J5
Conn_02x04_Counter_Clockwise

U9
PI4ULS5V202

Integrated 10K resistors on both sides.
VCCA can be lower, greater or equal to VCCB.

Fiducials

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
FID4 Fiducial
H1 MountingHole GND
H2 MountingHole GND
H3 MountingHole GND
H4 MountingHole GND

Maximilian Kern

Sheet: /
File: PicoLab.kicad_sch

Title: PicoLab

Size: A2	Date: 2026-08-22	Rev: 2
KiCad E.D.A. 10.0.3		Id: 1/1